BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 028 · Karnaugh maps 1: start with four squares

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Write −13 as an 8-bit two’s-complement integer .

Q2. For the circuit shown, simplify Q = ¬(A ∨ B) using De Morgan’s Law.

Q3. Complete one full left-to-right pass of ascending bubble sort. What is the resulting array?

Q4. Normalise this positive binary mantissa without changing its value. Give the new exponent.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — Karnaugh map

Recall the term — Group

Recall the term — Adjacent

Recall the term — Simplify

Recall the term — Gray code

New learning

Map the truth-table outputs — For each combination of A and B, find the same output Q in the map. The grid moves outputs; it does not change them.

Group adjacent 1s — Circle the two 1s in the bottom row. A group can contain 1, 2, 4, 8… adjacent 1s. Use the largest useful group.

Compare the input values — For the two circled cells: — CODE:
A = 1, B = 0
A = 1, B = 1

Keep only what stays the same — Look at each input variable across the grouped cells (where Q = 1): — In our group, A = 1 in both cells, while B changes. Keep A, leave out B: Q = A .

Why Gray code? — From 01 to 10 , two bits change — these cannot be neighbouring K-map columns. — Each neighbouring pair differs by one bit only , including the end pair 10 ↔ 00 .

Read the three-input layout — Two bits label each BC column. Moving sideways changes one input only. This allows useful groups in three-variable maps.

Worked example

How does this four-square map simplify?

1. Read the input combinations — Only the two bottom cells contain 1. They represent A=1, B=0 and A=1, B=1 .

2. Circle the adjacent 1s — Both outputs are 1 and the squares share an edge.

3. Compare each input

4. Write the simplified expression — Keep A because it remains 1. Remove B because it changes between 0 and 1. — Check: Q is 1 for both values of B whenever A = 1.

We Do · Practise with guidance

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Circle a vertical pair

Two 1s lie in the same column.

1. Identify the group

Circle the two adjacent 1s.

2. Verify the grouping

They form one vertical group of two.

3. What stays fixed?

For the circled cells, write A and B as 0 or 1. Which value changes?

4. Keep or remove

A: 0 → 1 (changes); B: 1 → 1 (always 1).

5. Write Q

Which input remains after removing the one that changes?

6. Explain the result

Circle a horizontal pair

Two 1s lie in the same row.

Circle both 1s in one horizontal pair.

The top row contains the pair.

State whether A and B are always 0, always 1, or changing.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

1. One horizontal pair
Circle both adjacent 1s. For A and B, state which stays constant or changes. Write the simplified Q.

Question 2

2. Just one 1
Which input values produce the only 1? Write Q as one AND term using A and B.

Question 3

3. Two pairs with an overlap
Circle two useful pairs, identify the fixed input in each, then combine the terms with OR.

Question 4

4. Explain Gray-code order
Complete the two missing column labels, then explain why 01 and 11 are adjacent and why the order 00, 01, 10, 11 is unsuitable for a K-map.

Question 5

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

5. Simplify a group of four
The columns use BC Gray-code order. Circle all four 1s, compare A/B/C, then write Q.

Question 6

6. Simplify a pair
Circle the two adjacent 1s. Which two inputs stay constant? Write the simplified Boolean term.

Question 7

7. Free-popcorn voucher
E = voucher shown, F = soft drink, G = water. Free popcorn is offered when E AND (F OR G). Circle two useful pairs and write R as an OR of two AND terms. Adapted from OCR Boolean Algebra Q2(b).

Exit Ticket · Answer before checking

A variable that changes value inside a Karnaugh-map group is removed from that group's simplified term.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 11110011 . Start with +13 = 00001101. Invert the bits → 11110010, then add 1 → 11110011.

Q2. Q = ¬A ∧ ¬B . NOT of an OR becomes AND of the individually negated inputs. Both expressions are equivalent.

Q3. [4, 6, 2, 8, 9] . Compare neighbouring items and swap where needed: [4,9,6,2,8] → [4,6,9,2,8] → [4,6,2,9,8] → [4,6,2,8,9].

Q4. 0.110100×2^2 . Shift the mantissa point two places right (0.001101 → 0.110100), so decrease the exponent from 4 to 2. The numerical value stays the same.

You Do pages · Independent questions

Q1. Circle: Both 1s in the top row make one pair.
Compare: A=0 in both cells. B changes from 0 to 1.
Simplify: A stays 0, so use NOT A (¬A). B changes, so remove B.
Check: Q is 1 whenever A is 0.

Q2. Circle: A single 1 at A=1, B=1 cannot join another output 1. A group of one is valid.
Compare: A and B are both fixed at 1.
Simplify: Keep both positive terms with AND.
Check: Q=1 only for A=1 and B=1.

Q3. Group 1: The top row has A=0 throughout; B changes. The term is ¬A.
Group 2: The right column has B=1 throughout; A changes. The term is B.
Overlap: The top-right 1 belongs to both groups; this is allowed.
Simplify: Combine the two terms with OR, giving ¬A ∨ B.

Q4. Ordinary binary: 00 → 01 → 10 → 11
Gray code: 00 → 01 → 11 → 10
Each neighbouring Gray-code label differs in exactly one bit.
Complete the labels: 00, 01 , 11, 10 .
Why 01 and 11 are neighbours: Only the first input bit changes (0 → 1); the second bit stays 1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Why ordinary binary order is unsuitable: 01 → 10 changes both input bits, so those columns are not adjacent in a K-map.
Edge connection: 10 ↔ 00 also changes only one bit; the left and right edges touch conceptually.

Q5. Circle: All four 1s form a 2×2 group across BC=01 and BC=11 and both A rows.
Compare: A changes 0 ↔ 1; B changes 0 ↔ 1; C remains 1.
Simplify: Remove A and B. Keep C.
Check: Q=1 whenever C=1, whatever A and B.

Q6. Circle: The two 1s are in row A=1, under BC=01 and BC=11.
Compare: A stays 1; C stays 1; B changes 0 → 1.
Simplify: Keep A and C joined by AND. Remove B.

Q7. Group 1: In row E=1, columns FG=01 and 11 give E ∧ G. F changes while E and G stay 1.
Group 2: In row E=1, columns FG=11 and 10 give E ∧ F. G changes while E and F stay 1.
Overlap: FG=11 is in both pairs and that is allowed.
Simplify: Join the two AND terms with OR. An equivalent factored form is E ∧ (F ∨ G).
Adapted classroom K-map based on the OCR Boolean Algebra popcorn scenario; not a reproduced OCR K-map.

Exit Ticket · Model answer

True. A variable that changes within a valid K-map group is eliminated from the simplified term.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.